

Machine Learning – Regression:

In Machine Learning Domain, under supervised learning for regression method,(prediction output based on numerical values), for the dataset,"50_Startups.csv", first created model using **Multiple Linear Regression** Algorithm, where R2_score is **0.9358**.

Further trained the model using **Support Vector Machine** Algorithm, with hyper tuning parameters, to check for the greater R_score value, which is listed below.

Hyper tuning Parameters	Kernel=Linear (R2_score)	Kernel=rbf (Radial Bias Function) (R2_score)	Kernel=Poly (R2_score)	Kernel=Sigmoid (R2_score)
C=10	-0.0396	-0.0568	-0.0536	-0.0547
C=100	0.1064	-0.0507	-0.0198	-0.0304
C=500	0.5928	-0.0243	0.1146	0.0705
C=1000	0.7802	0.0067	0.2661	0.1850
C=2000	0.8767	0.0675	0.4810	0.3970
C=3000	0.8956	0.1232	0.6370	0.5913

From the above table, the model performed well for **C=3000** and kernel =" **Linear**".

Next, created model using **Decision Tree** algorithm with the parameter as follows:

S.No.	Criterion	Splitter	R2_score
1.	squared_error	best	0.9268
2.	squared_error	random	0.7521
3.	absolute_error	best	0.9619
4.	absolute_error	random	0.8028
5.	poisson	best	0.9164
6.	poisson	random	0.9058

From the above table, the criterion=" **absolute_error**" with splitter=" **best**" performed well as it has the highest R2_score.

So far, comparing the algorithms Multiple Linear Regression, Support Vector Machine and Decision Tree for the dataset "50_Startups.csv", Decision Tree algorithm performed well, that is it created the best model with high accuracy score.